

Gastric Cancer Dx and Staging

1 Intro L 1

I'm Dr Jonathan Salo, I'm a GI Cancer Surgeon in Charlotte, North Carolina. These videos are designed to educate you about cancer and its treatment and help you and your cancer care team make the right decisions for you. .

2 Gastric Cancer 7 (overlay)

Our topic for this video is gastric cancer (2)

3 Gastric Cancer

To start, gastric cancer is cancer of the stomach. (3) The terms can be used interchangeably

4 Torso 2

When we think about digestion, food passes from the throat into the esophagus (4)

5 Stomach Zoom

From the esophagus it passes into the stomach. After leaving the stomach, it passes into the duodenum and finally into the jejunum, the first part of the small intestines. (5)

6 Stomach Right

The focus in this video is on cancers of the stomach. (6)

7 GEJ and Stomach

As it turns out, cancers can also form in the esophagus. (7)

8 Eso GEJ Stomach

To make things more complicated, cancers can also form in the area where the esophagus and stomach meet, called the gastroesophageal junction. (8)

9 Eso GEJ

The cancers of the gastroesophageal junction behave similar to cancers of the esophagus, so they are often grouped together.

We have other videos that focus on cancer of the esophagus and gastroesophageal junction

10 Stomach

This video focuses on cancers of the stomach, or gastric cancers.

11 Tumor 7 (overlay)

Let's review some cancer terms. A **tumor** is an abnormal growth. (11)

Tumors can be either benign or malignant

12 Tumor

A **benign** tumor of the stomach may grow over time, but it won't ever spread anywhere else

13 Tumor

A **malignant** tumor of the stomach has the *potential* to spread elsewhere in the body. This could be to the lymph nodes, the lungs, the liver, so the bone *or* spread inside the abdominal cavity..(13)

14 Tumor

Cancer is another term for a malignant tumor (14)

15 Upper Endoscopy = EGD. 2

In most cases, the diagnosis of esophageal cancer is made by endoscopy, also called and EGD for esophago-gastroduodenoscopy (15)

Under sedation, a flexible scope is passed through the mouth into the stomach, which allows viewing the inside of the stomach.

If a tumor is found in the stomach, a small portion can be removed, which is called a **biopsy**

16 (Microscope)

The biopsy is examined by a pathologist, who will determine whether the tumor is benign or whether it is malignant, meaning a cancer (16)

17 Staging. 7 Overlay

If the biopsy shows cancer, the next step is staging. Staging is the process of finding out the *size* of the tumor and whether or not there has been spread to the lymph nodes or other places in the body.(17)

18 Staging

The cancer treatment is determined by the stage (18)

Once the stage has been determined, it will be possible to determine the best therapy

19 Staging

Staging is going to focus on three areas: (19)

The first is the *tumor*. How large is the tumor? And more importantly, how deeply has it grown into the wall of the esophagus?

The second is whether there is spread to the *lymph nodes*.

The third is whether there is spread to other parts of the body

20 Staging TNM

T for Tumor N for Lymph Nodes

...spread to other parts of the body is called *metastasis* (20)

21 Torso. 2 Slides Only

We will start by looking at the wall of the stomach. The wall of the stomach has multiple layers, shown here.

(pause for “video”) 22-23-24-25-26-27-28-29-30-31

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32 Tumor Cross Section 2

The inner layer is the mucosa, (32)

followed by the submucosa,

followed by the muscularis, or muscle layer

Outside of the muscularis is the serosa.

Surrounding the stomach are lymph nodes. The purpose of lymph nodes is to filter the blood and help fight infections, but in some cases cancers from the stomach can spread to the lymph nodes

33 Early Tumor

In its earliest stages, cancer of the stomach start in the inner, or most superficial layer, called with mucosa. (33)

34 T1b Tumor

With time, however, the cancer can continue to grow and invade deeper into the wall of the esophagus. (34)

35 T2 Tumor

(pause) - (35)

36 T3 Tumor

(pause (36)

37 Nodes

The deeper the cancer invades into the wall of the esophagus, the more likely it is that cancer cells can spread to lymph nodes (37)

38 Staging 7

To review, the stage consists of 3 parts: (38)

- T for Tumor
- N for Nodes
- M for Metastasis or spread to other organs

39 Staging T

We will first consider the tumor (39)

40 T1a

T1a tumors invade just the top layer, or mucosa (40)

41 T1b

T1b tumors invade into the submucosa (41)

42 T2

T2 tumors invade into the muscularis (42)

43 T3

T3 tumors go all the way through the muscularis (43)

It can be difficult to tell how deep a tumor has invaded just by looking from the surface.

44 EUS 7 Overlay

For small tumors, *endoscopic ultrasound* can be helpful. This is an endoscopic procedure done under sedation much like a EGD and it can help determine the T stage of a stomach tumor. (44)

45 EUS Drawing 2 Slides

The EGD scope has an ultrasound probe on the end, which can be used to examine the tumor (45)

46 Staging 2 Overlay

Evaluation of the *lymph nodes* is the second part of staging (46)

47 N0

Cancers with no signs of spread to the lymph nodes are considered “N-zero” (47)
...while cancers with signs of spread to the lymph nodes are considered “N-positive”

48 CT Scan

A CT scan is usually the first test to look for enlarged lymph nodes. (48)

49 CT Scan

(pause)

50 PET Scan 7

A PET scan is similar to a CT scan. It differs because of a tracer which is injected intravenously which lights up areas of cancer (50)

51 PET Images 2

In this example of a PET scan, you can see how the area of the cancer lights up in yellow (51 - PET image)

52 Staging M **7* Overlay

The third part of staging is the M stage, for metastasis. (52)

53 Staging M0

Cancers that had not spread to other organs are considered M-zero (53)

54 Staging M1

Cancers that have spread to other organs are considered M1 (54)

55 Liver 2 Slides

There are several common sites to which gastric cancer can spread, such as the liver (55)

56 Lungs

... lungs (56)

57 Bone

... or bones (57). The CT scan and PET scans are used to make sure there are no signs of spread.

58 Carcinomatosis

An additional pattern of potential spread is *carcinomatosis* (58). This is spread inside the abdominal cavity.

59 Carcinomatosis 7 Overlay

Cancer can grow in small nodules no larger than a grain of rice, and frequently can't be seen on CT or PET scans. (59)

60 Carcinomatosis - Laparoscopy

Detection of carcinomatosis requires looking inside the abdominal cavity with a small surgical procedure called *laparoscopy* (60)

61 Laparoscopy

Laparoscopy is a minor surgical procedure performed under a general anesthetic as an outpatient

62 Laparoscopy

Several small incisions are made, and a scope is inserted to examine the inside of the abdomen

63 Laparoscopy image 2

A laparoscope is a telescope the diameter of a pencil (63)

64 Laparoscopy Torso

A laparoscope can be used to inspect the inside of the abdomen through a small incision (64)

65 Staging 7

To review, staging consists of three parts: T for Tumor N for Nodes and M for Metastasis (65)

66 Treatment

We will briefly summarize treatments:

Superficial cancers are T1 and limited to the mucosa. These can usually be treated with endoscopic therapy alone. Surgery can be avoided in many of these cases. (66)

67 Localized

Localized cancers are T2 tumors that invade the muscle and are treated with surgery, also termed esophagectomy (67)

68 Locally Advanced

Locally-advanced cancers are T3 or Node-positive and are usually treated with chemotherapy first, followed by surgery (68)

69 Metastatic

Metastatic cancers are treated initially with intravenous chemotherapy. (69)

70 Metastatic

This allows cancer-killing drugs to circulate through the bloodstream (70)

71 Metastatic Additional Therapy

After initial treatment with chemotherapy, some patients are candidates for additional therapy (71)

72 Metastatic Surgery

Surgery is used in some patients with metastatic disease after initial chemotherapy. (72)

73 Metastatic Gastrectomy

Gastrectomy can be used to remove the tumor in the stomach (73)

74 Metastatic Cytoreductive

Cytoreductive surgery can be used to remove areas of spread of cancer within the abdomen (74)

75 Metastatic IP Chemotherapy

In some patients, chemotherapy can be administered into the abdominal cavity, also known as *intraperitoneal* chemotherapy (75)

76 HIPEC

Intraperitoneal chemotherapy can be administered as a surgical procedure called *HIPEC* (76)

77 IP Catheter

Intraperitoneal chemotherapy can also be administered as an outpatient after surgical placement of a intraperitoneal catheter (77)

78 Talking Head 1 Talking Head

We are working on preparing videos to address each of these different treatments, so please subscribe to be notified when we post new videos.

(pause)

We hope you have found this video helpful. This videos and others like it are designed to educate patients and families about esophageal cancer

Feel free to leave a comment about your experience with gastric cancer or if you have suggestions for new videos.